

PHILOSOPHY OF MATH

1. MATHEMATICAL PARADOXES

1.1. **Russel's Paradox.** Suppose there is a town with a exactly one barber who shaves only those people who do not shave themselves. Does the barber shave himself?

Thinking through all the possibilities, we arrive at a contradiction.

- If the barber shaves himself, then we arrive at a contradiction (why?)!
- But if he does not shave himself, we also arrive at a contradiction (why?)!

So, it is impossible to answer the simple question: does the barber shave himself?

This is a variation of Russel's paradox: Suppose X is a collection of items which contains all collections of items that do not contain X . Does X contain itself?

This paradox was used to show that the foundation of mathematics used by prominent mathematicians, such as Georg Cantor, contained contradictions.

1.2. **A solution to Russel's paradox.** In the early twentieth century, Ernst Zermelo and Abraham Fraenkel proposed **Zermelo-Fraenkel set theory**, an axiomatization of the foundations of mathematics that aimed to rid of paradoxes like Russel's paradox.

In modern mathematics, **Zermelo-Fraenkel set theory with choice** is the most commonly accepted axiomatization of set theory.

But the **axiom of choice** is (somewhat) controversial¹, and it itself introduces new paradoxes!

1.3. **What is the axiom of choice? Finite case:** Suppose there are 10 buckets, each filled with random pieces of candy. The axiom of choice says it is possible to choose exactly one piece of candy from each box.

Infinite case (countable): Suppose there are infinitely many buckets, each filled with random pieces of candy. Assume you can still count the buckets, so that if you lived for infinity, you could keep counting the buckets, 1,2,3,4,5,6,7,... The axiom of choice says that it is *still* possible to choose exactly one piece of candy from each box.

Infinite case (uncountable): Suppose there are infinitely many buckets, each filled with random pieces of candy. This time, however, even if you lived for infinity, there are so many buckets that you could never count them all: if you pointed to one bucket and then another, there would always be another one in between them (think of the decimals between 0 and 1 – you could never list them all!). The axiom of choice says that it is *still* possible to choose exactly one piece of candy from each box.

Nevertheless, much of modern mathematics relies on the uncountable axiom of choice, and most mathematicians don't really think about its seeming impossibility. But it does come with its own paradoxes!

¹Most practicing mathematicians use the axiom of choice, but this is still fun to think about.

1.4. **The Banach-Tarski paradox.** The Banach-Tarski paradox is equivalent to the axiom of choice. That is, the Banach-Tarski paradox exists if and only if we accept the axiom of choice.

The Banach-Tarski Paradox says that given a solid ball, it is possible to break the ball into finitely many pieces, then reassemble those pieces to form **two solid balls, each identical to the first.**

Unlike Russel's paradox, the Banach-Tarski paradox is not mathematical or logically incorrect; instead, it contradicts what we think is *physically* possible – indeed, we cannot actually cut up a bouncy ball and reassemble the pieces to get two identical copies of the original ball.

It turns out that many important and very useful mathematical results are equivalent to the axiom of choice. This is why most mathematicians choose to not think about the contradictions and paradoxes that come with accepting it.

2. IS MATH UNIVERSAL?

If different parts of the world could not communicate, would math be the same everywhere? How can different cultures and environments affect the way people do math?

2.1. **Law of the Excluded middle.** As an example, some mathematicians, called **constructivists**, reject the so called “Law of the Excluded Middle.” That is, they believe that just because a statement is not true, it does not mean it is false, and vice versa: if a statement is not false, it does not mean it is true.

Some constructivists aim to obtain all modern mathematical results without the use of the law of the excluded middle. Do you think this is possible?

3. IS MATH INVENTED OR DISCOVERED?

Many prominent mathematicians throughout history have subscribed to the idea that math is discovered, not invented. Here are some quotes:

- “I have myself always thought of a mathematician as in the first instance an observer, a man who gazes at a distant range of mountains and notes down his observations. His object is simply to distinguish clearly and notify to others as many different peaks as he can.” – G.H. Hardy
- “I am not qualified to say whether or not God exists. I kind of doubt He does. Nevertheless I'm always saying that the SF (The SF is the supreme Fascist, the Number-One guy up there) has this transfinite book-transfinite being a concept in mathematics that is larger than infinite-that contains the best proofs of all mathematical theorems, proofs that are elegant and perfect.” – Paul Erdős

Others, however, thought differently:

- “God created the natural numbers, all else is the work of man.” – Leopold Kronecker
- “Mathematics is a game played according to certain simple rules with meaningless marks on paper.” – David Hilbert